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Correlation of magnetic probe and neutron signals with interferometry figures on the plasma focus discharge

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Abstract
In this paper the results of temporally resolved measurements using calibrated azimuthal and axial magnetic probes are presented, together with interferometry and neutron diagnostics performed on the PF-1000 (IPPLM, Warsaw, 2 MA) device with a deuterium filling and 10^{11} neutron yield. The probes located in the anode front at three different radial positions allow determination of the dominant part of the discharge current flows behind the imploding dense plasma layer. The current sheath is composed of both the axial and azimuthal components of the magnetic field. After reaching the minimum diameter, the current sheath continues in a radial motion to the axis and then penetrates into the dense plasma column. At the final phase of stagnation, the dominant current passes through the dense column. The probes located on the axis of the anode front registered an increase and a decrease in the pulse of the axial component of the magnetic field in correlation with the formation and decay of the dense plasmoidal structure. The estimated values of the axial component of the magnetic field at the center of the plasmoids in the first neutron pulse and close before its decay and dominant neutron production can reach 2 and 10 T; it is 10–30% of the value of the azimuthal magnetic field of the dense column boundary.

1. Introduction

A plasma focus device with a deuterium filling is an efficient source of neutrons produced by fast deuterons through the beam–target mechanism. Non-thermal mechanisms of acceleration of deuterons up to an energy of hundreds of keV are under discussion [1–5], in connection with the evolution of instabilities, microturbulences, filamentary structures and transformations of magnetic fields. Recently, in the PF-1000 device, a plasma has been investigated with laser interferometry, x-ray and neutron diagnostics [6]. The interferometric diagnostics showed several toroidal, helical and plasmoidal structures inside the plasma column. They were usually observed during all phases of pinch evolution: the imploding current sheath, the stagnation and evolution of instabilities. Intense high-energy electrons and ions were produced in correlation with the formation and disintegration of the plasmoids and constrictions. These structures contain closed currents, and their evolution can be explained in terms of the spontaneous transformation of the magnetic field topology performed by the reconnection of magnetic lines [7, 8].

Experimental measurements of magnetic fields in plasma foci are very rare. For example, in [9] the megagauss azimuthal component of the magnetic field was estimated using the Zeeman effect and in [10, 11] the value of the magnetic field and the distribution of the current during the implosion phase in operation with different filling gases on the PF 3 device (Kurchatov Institute, Moscow) were presented, and the closed axial current with return component was observed with the help of the magnetic probe technique. Recently, measurement using magnetic probes has been performed on the PF-1000 device. It demonstrated only a small perturbation of the plasma with the body of the probe and a long lifetime of the registration of a reliable signal before the probe's destruction in the majority of shots. The conclusions from the measurements of the azimuthal component of the magnetic field and the distribution of the discharge current passing the probe positioned in the anode face at the radial position of 4 cm were published in [12]. Namely, the current in the pinch phase passing through the radius below 4 cm can achieve 80–100% of the total discharge current in shots with a high neutron yield. In 2011, an attempt was made to measure the axial magnetic field at the PF-1000 facility with the help of magnetic probes [13]. The axial component of the magnetic field was observed at the stage of plasma current sheath propagation toward the axis. However, these results were difficult to interpret unambiguously because of the possible strong influence of the azimuthal component of the magnetic field on the probe signals. In [14] the existence of the axial component of the magnetic field was improved in the imploding current sheath and pinched column. Its presence is an important argument for the existence of closed magnetic structures, which could play a role in the mechanisms of deuteron acceleration. In [14] the uncertainty of the value of the magnetic field was considered as well. The coil that registered the axial component also partially registered the azimuthal component. For this reason, the mutual influence of azimuthal and axial components was measured at the calibration and this influence was considered during the processing of the signals. The limitation of the reliability of the presented results was caused by the poor statistics of only a few shots existing for each position of the probe. The small values of the axial component obtained, usually below 1 T, were understood as a consequence of the position of the probe coil at the anode face, often far away from the transformed structures. Therefore, the value of the magnetic field in those structures can be higher.

In this paper, we present the results of measurements using absolute calibrated magnetic probes described in [14] in temporal correlation with interferometry images and neutron signals obtained from the PF-1000 facility operating with deuterium. In section 2, we describe experimental devices and diagnostics. In section 3, the correlation of the signals of magnetic probes with images is described during plasma implosion and neutron production. In section 4, we summarize the results and formulate our conclusions.

2. Device and diagnostics

The measurements described here were performed within the PF-1000 facility equipped with Mather-type coaxial electrodes of 480 mm length. The cathode of diameter 400 mm was composed of twelve 82 mm-o.d. stainless-steel rods distributed symmetrically around the anode circumference. The condenser bank was charged to a voltage of 24 kV, which corresponded to a discharge energy of 384 kJ. The discharge current during the pinch phase amounted to 1.5–1.8 MA. The initial pressure of the pure deuterium filling was 240 Pa and the neutron yield was in the level of 10^{11} per shot.

The signals of the voltage, the current derivative in time and the current were measured within the current collector

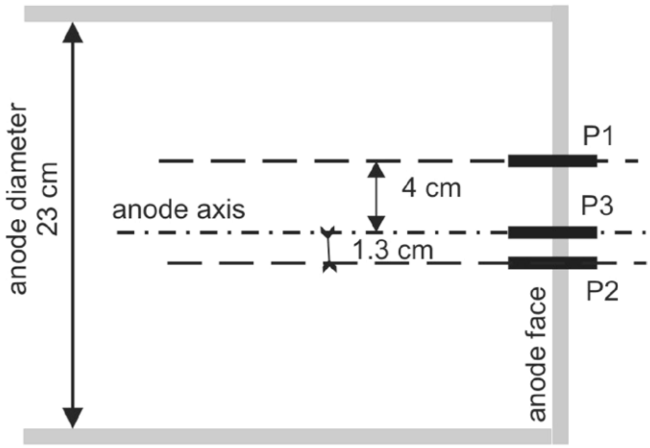

Figure 1. Schematic diagram of the position of magnetic probes in the anode face. Both dBz and d$B\varphi$ coils were positioned at distances 0–5 mm in front of the anode face.

near the insulator at the bottom of the anode. Soft x-ray (SXR) pulses in the range of photon energy of 0.6–15 keV were recorded with a silicon PIN detector shielded by means of a Be-foil of 10 μm thickness. To determine the time of generation of hard x-rays and neutrons by the time-of-flight method, we used a scintillation detector coupled with a fast photomultiplier situated side-on at a distance of 7 m from the z-axis. The interferometric measurements were performed with a Nd : YLF laser operated at the second harmonics (527 nm). The laser pulse (lasting less than 1 ns) was split by a set of mirrors into eight separated beams, which passed through a Mach–Zehnder interferometer. These beams investigated the plasma region with a mutual delay of 30 ns ranging from 0 to 210 ns. The total neutron yield was calculated from the data recorded with four silver-activation counters. The minimum of the current derivative was assigned as $t = 0$.

The azimuthal and axial components of the magnetic field were recorded with probes placed in the anode face at radii of 4, 1.3 and 0 cm in positions perpendicular to the laser diagnostic beam. The method of usage of azimuthal probes was described in [12]. Each probe measured the time derivative of the azimuthal component of inductance of the magnetic field d$B\varphi/\mathrm{d}t$ (in brief, derivative of the azimuthal magnetic field, d$B\varphi$) at its location. To determine the current flowing within the radius at which the probe was installed, the signal was integrated numerically under the assumption that the distribution of the plasma current was symmetrical with respect to the system axis. The probes were calibrated individually and this process is described in detail in [12]. The calibration accuracy was better than 5%, and the accuracy in determining the magnetic induction in the plasma due to axial symmetry with account of calibration was about 15–20%.

During the lateral experimental campaign, together with the azimuthal, the axial magnetic probes were used, and the results obtained were described in [14]. The coil 2 mm in diameter was positioned in the probe body on the level of the anode face at a radial distance of 4 and 0 cm. A detailed description of the probes was published in [14] and a schematic diagram is shown in figure 1. Each probe coil was calibrated

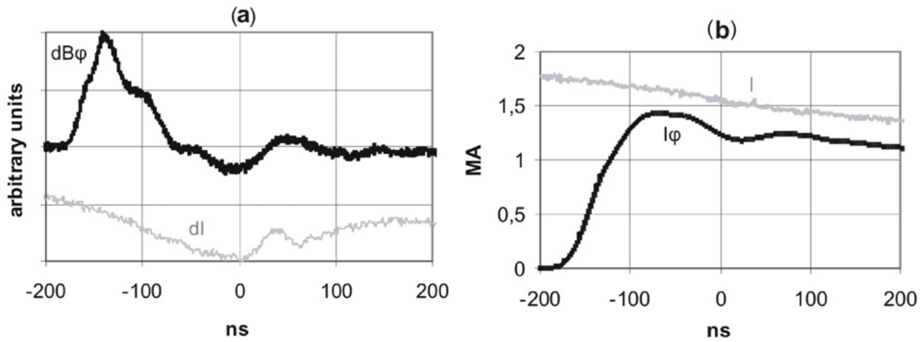

Figure 2. Shot #9340: (*a*) signals of magnetic probes d$B\varphi$ (black) at 4 cm and of the current derivative dI (gray), (*b*) temporal dependence of the total current measured with the Rogowski coil I (gray) and the integral value of the total current $I\varphi$ (black) calculated for $B\varphi$ registered at 4 cm.

for its sensitivity to the azimuthal magnetic field and this influence was subtracted during signal processing to obtain the pure dBz/dt (dBz) signal. The uncertainty of dBz signals, including the influence of calibration and subtraction of $B\varphi$, was estimated up to 50%. The temporal response of the magnetic field penetration into the probe was estimated as 1 ns.

The images of laser interferometry demonstrated relatively small perturbations of the dense plasma introduced by the probes, which did not practically affect the process of pinch formation and evolution. The lifetime of the coil in the probe before its destruction was different in dependence on the configuration of the plasma around the probe and the radial position of the probe. The feedback of the reliability of the probe signal was verified by configuration of the plasma in interferograms, comparison of signals from both coils and the state of the probes after the shot. In some cases we were able to use the probe in another shot.

3. Experimental results

3.1. Signals of magnetic probes at an axial distance of 4 cm from the axis

3.1.1. Position of the current sheath toward the imploded dense plasma layer. Magnetic probes positioned at a radial distance of 4 cm from the axis registered d$B\varphi$ during the implosion of the dense plasma layer. In this configuration, we realized eight shots. On average, d$B\varphi$ started at -200 ns, its maximum was reached at -160 ns and the maximum value of $B\varphi$ (d$B\phi = 0$) was achieved at -110 ns with the standard deviation of 10 ns from shot to shot. An example of d$B\varphi$ signals and the current derivative for shot #9340 is shown in figure 2(*a*). The total current measured with the Rogowski coil and the integral value of the total current calculated for the registered $B\varphi$ are shown in figure 2(*b*). The results, in which up to $80 \pm 20\%$ of the total discharge current passed the diameter below 4 cm, conformed to the results recently obtained from the PF-1000 device [11].

In shot #9340 in figure 2, the signal of the azimuthal field started at -170 ns and reached its maximum at a current of 1.4 MA at -71 ns. Later, its amplitude slowly began to decrease. From the interferograms registered in figure 3 at -139 and -79 ns, we can estimate the position of the current layer near the anode toward the imploding dense plasma

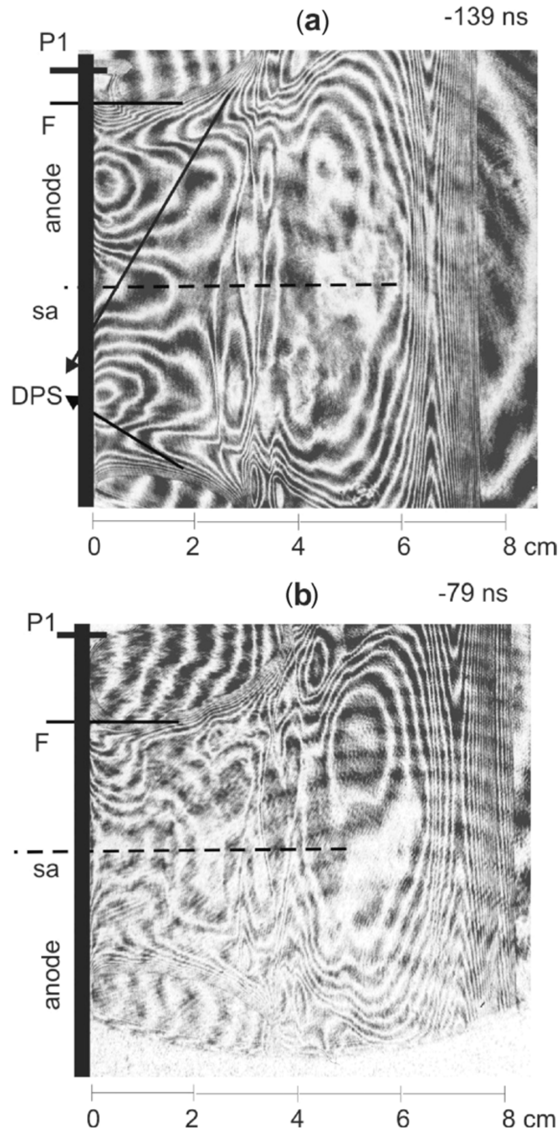

Figure 3. Shot #9340: interferograms registered at -139 ns (maximum of d$B\varphi$) and -79 ns (maximum of $B\varphi$) with labeled positions of anode face (black streak) and anode axis (sa). $P1$ labels the position of the azimuthal magnetic probe located at 4 cm, and F the position of the front of the current sheath and DPS the position of the dense plasma layer.

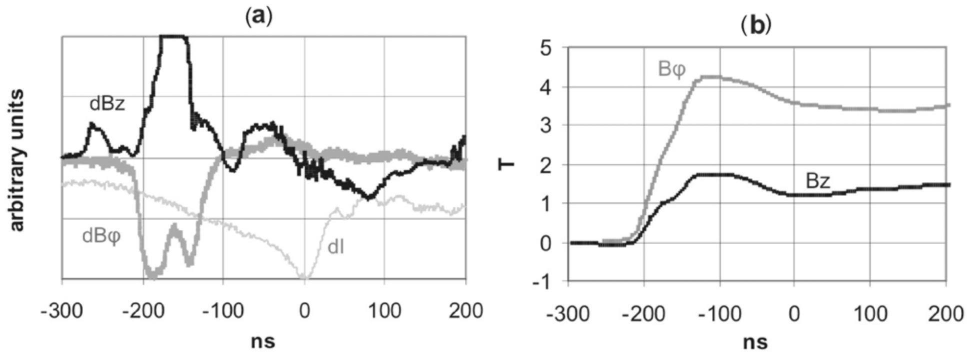

Figure 4. Shot #9344: (*a*) temporal evolution of the current derivative dI (thin gray), the signal of the azimuthal d$B\varphi$ (thick gray) and axial dBz (black) components of the magnetic field registered at 4 cm distance, (*b*) temporal evolution of the azimuthal $B\varphi$ (gray) and axial Bz (black) components of the magnetic field.

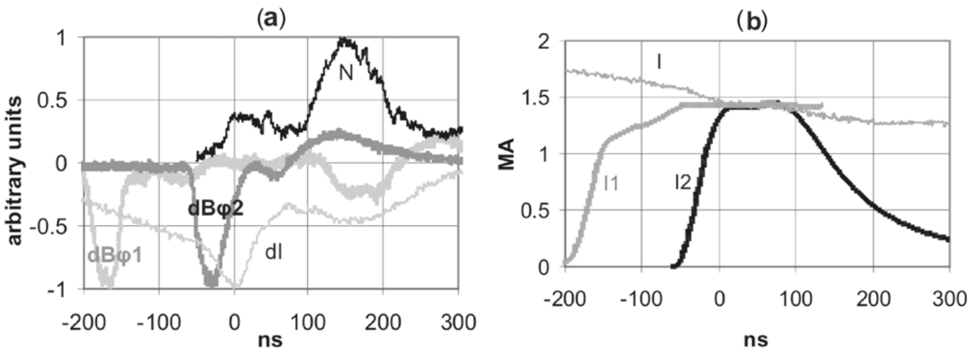

Figure 5. Shot #9367: (*a*) signals of current derivative dI (shadow, thin), neutrons N (black thin; the waveform of neutrons registered side-on was shifted to the position of their start between electrodes supposing an energy of 2.45 MeV), the azimuthal magnetic probe at 4 cm d$B\varphi$1 (pale shadow, thick) and the magnetic probe at 1.3 cm d$B\varphi$2 (dark shadow, thick), (*b*) current flowing inside 4 cm radius I1 (shadow, thick) and 1.3 cm radius I2 (thick dark shadow) calculated from d$B\varphi$ signals in comparison with the current registered with the Rogowski coil I (thin).

layer, knowing the velocity of implosion. The mean value of this velocity (2.1×10^5 m s^{-1}) with 25% standard deviation from shot to shot was estimated from the velocity of the imploded dense plasma layer calculated from its position in the interferograms registered at different times. The current sheath is positioned behind the dense plasma layer, and its position about 10 ns before its end is labeled with the position of probe P1 in figure 3(*b*). The full-width of the current layer is 1.6–2.6 cm.

3.1.2. Correlation of Bz with Bφ. In three shots, we used double magnetic probes for measurement of d$B\varphi$ and dBz at 4 cm distance. Then, we were able to compare the radial distribution of the axial and azimuthal components of the magnetic field. The waveforms of d$B\varphi$ corresponded to the waveforms described in the above paragraph. dBz signals differ in the first small prepulse preceding the current sheath, starting 70–90 ns before the main pulse and with an orientation of this Bz field directed from the anode to the cathode. The start and finish of the main pulse dBz (with amplitude partially limited) correlated with d$B\varphi$. In the calculated values of Bz in figure 3(*b*), the influence of the $B\varphi$ signal registered with the axial probe coil using the above-mentioned calibration of its sensitivity to the azimuthal magnetic field was subtracted according to the formulae presented in [14]. The sign of the Bz component was opposite to the first small pulse and its amplitude was 2–8 times lower than $B\varphi$. In shot #9344 (figure 4), the maximal value of $B\varphi$ is roughly 4 T and Bz below 2 T. The prepulse dBz corresponds to the layer of 0.5 cm wide moving ahead of the imploding dense plasma layer (see in figures 3(*a*), 3(*b*), 6(*a*), 8(*a*) and 8(*b*)). This layer is characterized with a lower plasma density, below 10^{24} m^{-3}, inside of which the toroidal or helical-like structures are formed, in different shots at different z distances. The prepulse and the same signs of Bz were measured in the other two shots provided at this probe position.

3.2. Signals of magnetic probes d$B\varphi$ *at an axial distance of 1.3 cm*

Five shots were realized with two double magnetic probes d$B\varphi$ placed at the radial distances of 4 and 1.3 cm. The mean value of the start of the signals at 1.3 cm was −55 ns and the maximum was reached at 16 ns with a fluctuation of 5 ns. As an example, we chose shot #9367 with a neutron yield of 2.5×10^{10}.

The registered signals and images from shot #9367 are shown in figures 5 and 6. In figure 5(*a*) the waveform of neutrons registered side-on at 7 m distance was shifted to the position of its start between the electrodes supposing an energy of 2.45 MeV. Then, the HXR signal was positioned

300 ns before its origin. For this reason, it was removed from the neutron signal. The duration of the passing of the current sheath through probe P1 at 4 cm was 90 ns (the same as in figure 4(*a*)), and it was 70 ns for probe P2 at 1.3 cm. The position of the front of the d$B\varphi$ pulse in figure 5(*a*) correlates with the front of the dense plasma layer P2 shown in figure 6(*a*) at −68 ns; the maximal current density is positioned at probe P2 in figure 6(*b*) at −38 ns and the maximal current in figure 6(*c*) at +22 ns. At 22 ns (onset of the column's radial expansion near the anode) practically the entire value of the current passes through the dense plasma column. Therefore, after stopping the plasma implosion, the current sheath continued in radial motion into the dense column. Its penetration was completed during the radial expansion of the dense column near the anode, together with the ending of the first neutron pulse. This transport of the current to the axis can increase the inductance of the discharge current after reaching the minimal pinch diameter. $B\varphi$ calculated on the surface of the dense column with a radius of 1.3 cm in this shot at 22 ns has a value of 21 ± 4 T. The decrease in the current after 100 ns (figure 5(*b*)) correlates with the expansion of the radius of the dense column above 1.3 cm.

Unfortunately, in this position of $r = 1.3$ cm we did not use the coil for registration of dBz signals, so we had no evidence concerning Bz on the boundary of the dense column.

3.3. Signals of magnetic probes registered at the anode axis ($r = 0$ cm)

In this position of the magnetic probe, we registered four shots with signals of dBz and d$B\varphi$. The d$B\varphi$ signal registered both the azimuthal and radial components of the magnetic field only at the non-symmetry of the current distribution. Its value was considerably smaller than dBz. The dBz signals differed from shot to shot. The signal maximum in some shots reached 0.3–0.6 T (#9354, #9356 and #9358) while in shot #9357 it was 1 and 7 T. These different values can depend on the different distance of structures observed in the plasma column from the probe, as we discuss below.

For a detailed description, shot #9354 (NY 2.4×10^{10}) was chosen, where the pulses of dBz can be compared with the registered interferograms and neutron production. The high axial symmetry of interferograms in this shot enabled the use of the Abel inversion and the construction of densitograms. For better imagination and comparison of radial symmetry we show in figures both the original interfograms and densitograms.

In figure 7, we see the signals of the current derivative in time, neutrons, dBz and Bz. The signal dBz is composed of five different pulses. Unfortunately, two of them are limited. The first pulse was emitted during the implosion of the current sheath (−110–−30 ns), the second, limited, during the first neutron pulse (−20–0 ns), the third, huge, negative and limited, during the formation of the dense disk and plasmoid (0–80 ns), the fourth (80–140 ns) during the evolution of the constriction and the fifth, negative, after 140 ns, was emitted during the production of the main neutron pulse. The value of the axial component of the magnetic field in figure 7(*b*) for shot #9354

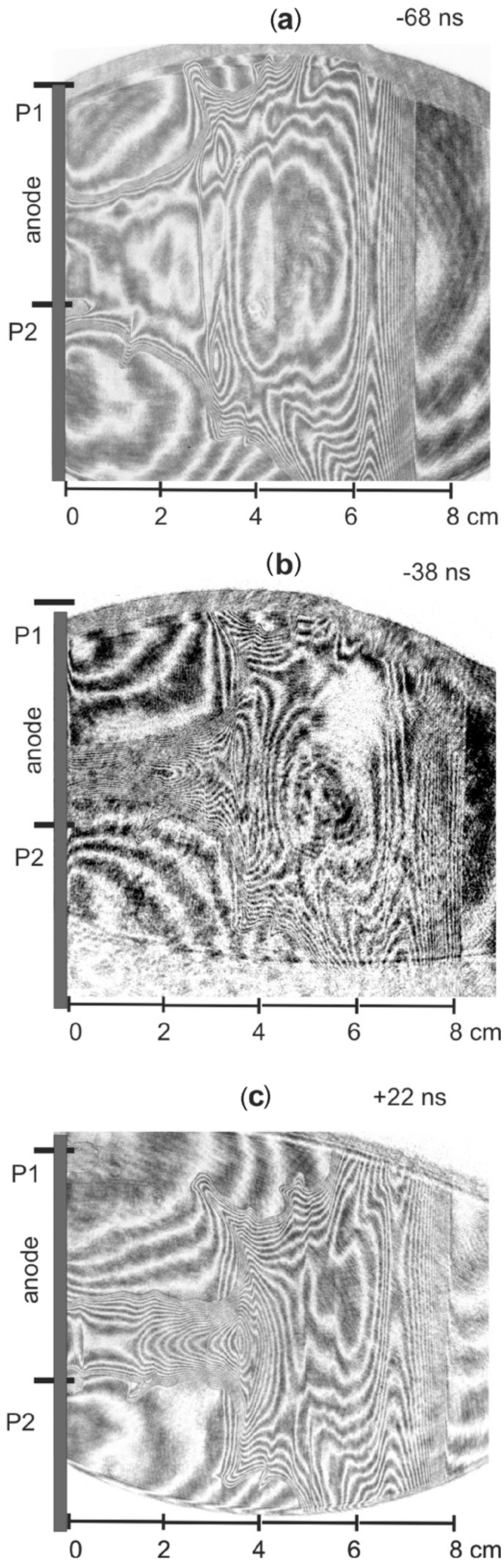

Figure 6. Shot #9367: interferograms registered at (*a*) −68, (*b*) −38 and (*c*) +22 ns. P1 (P2) is the position of the magnetic probe at 4 (1.3) cm radial distance.

was calculated, after −15 ns, from limited signals. Therefore, the temporal position of the local extremes of Bz is correct, but the value of the signal decrease and increase between the local extremes could be higher, and the amplitudes should be different. The character of the total Bz in this shot can be

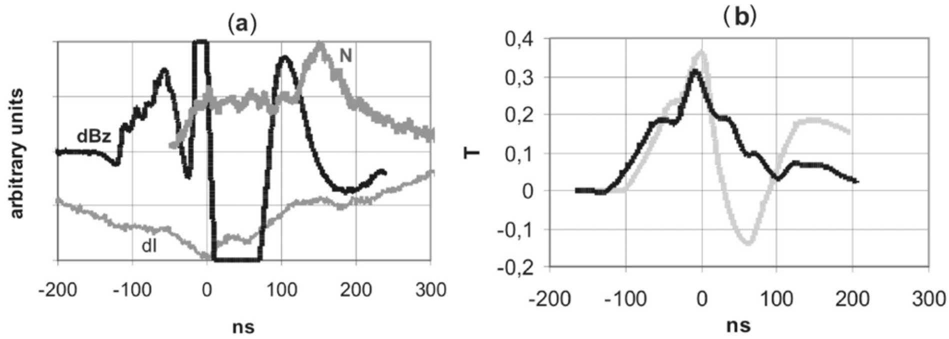

Figure 7. (*a*) Shot #9354: signals of current derivative dI (gray, thin), neutrons N (gray, thick; waveform of neutrons registered side-on was shifted to the position of their start between electrodes supposing an energy of 2.45 MeV) and signals of the dBz (black, thick), (*b*) Bz signals in shot #9354 (black) and in shot #9356 (shadow).

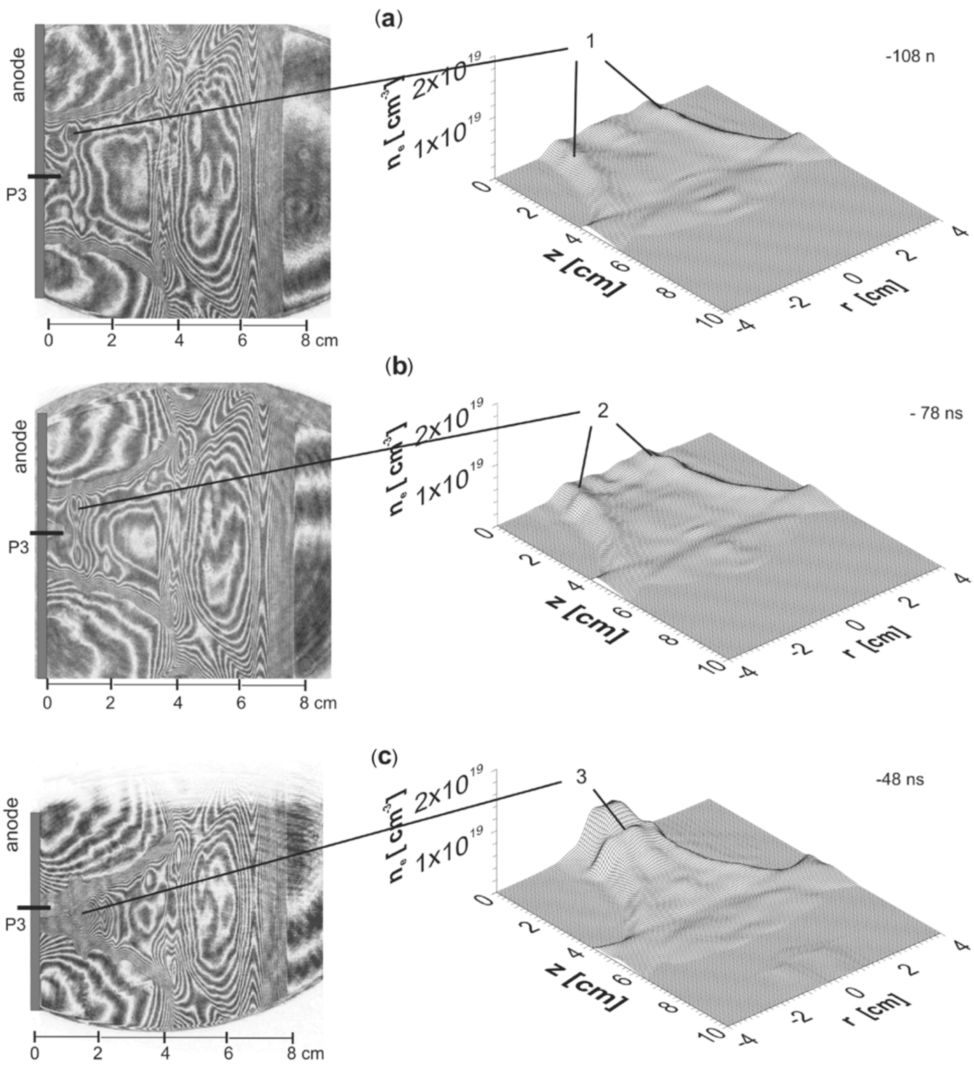

Figure 8. Shot #9354: interferograms and densitograms registered during the plasma sheath implosion at (*a*) −108 ns, (*b*) −78 ns and (*c*) −48 ns. Position of probe P3: $z = 0$ and $r = 0$.

estimated from comparison with a similar shot, #9356, with unlimited signals (in figure 7(*b*) in the shadow trace) where at 60 ns the value of −0.1−−0.2 T was registered.

In what follows, we describe the correlation of the individual dBz peaks with transformations of the structures imaged in interferograms during the implosion of the plasma layer and during neutron production.

3.3.1. Implosion of the dense plasma layer before the first neutron pulse. In figure 8, the interferograms showing the first dBz pulse were registered at times of −108, −78 and −48 ns. At –120 ns, the registration of the Bz pulse with the probe started (figures 7 and 8(*a*)) and at −30 ns it reached 0.2 T. This pulse corresponds to the small dBz prepulse described in section 3.1.2 and imaged in figure 4(*a*). As mentioned

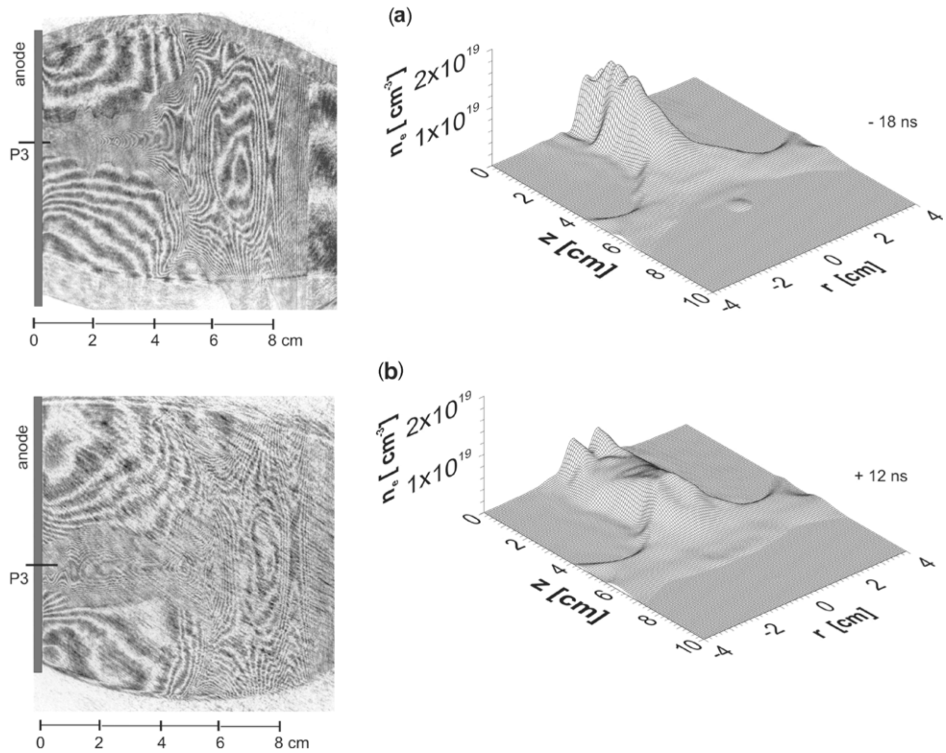

Figure 9. Shot #9354: interferograms and densitograms obtained during the first x-ray and neutron pulses at (*a*) −18 ns and (*b*) +12 ns. P3 is position of the dBz probe in $r = 0$.

above, it could be bounded with the layer moving in front of the imploding dense plasma sheath. In this layer, at about −108 ns at a z distance of 1–2 cm in front of the anode, the toroidal- or helical-like structures (1) started to form. We can assume the toroidal and plasmoidal structures as the sources of the azimuthal current and axial magnetic field. The Bz signal increases between −108 and −48 ns at the structure implosion. In figure 8(*b*), we see these toroidal-like structures (2) more clearly expressed. In figure 8(*c*), these structures are transformed into a disk-like shape with maximum density on the axis (labeled 3). The enhanced diameter of the pinched column in the z-position of this toroid and disk should be a consequence of the repelling pressure of the Bz magnetic field. Bz on the axis of the toroid can be calculated using the Biot–Savart law. For simplification, we can suppose the total azimuthal component of the current flowing through the narrow toroid with the radius of maximal density r. Then, the magnetic field Bzc at the center of this toroid at −48 ns can be estimated according to the formula $Bz = Bzc \cdot (\sin\varphi)^3$, where $\mathrm{tg}\varphi = r/z$. For distance $z = 1.0$ cm of the probe from the disk, radius $r = 0.8$ cm of the disk and $Bz = 0.2$ T at −48 ns, we can estimate the total magnetic field at the center of the toroidal structure to be about 1.2 T. In comparison with $B\varphi$ of about 15–25 T calculated for the surface of the dense plasma column Bz reached roughly 5–8%. At −18 ns, when the toroid is transformed into the disk-like structure, the radius decreased to 1.0 cm, the Bz reached 0.3 T and the magnetic field at its center can be estimated to be 2 T.

3.3.2. The first neutron pulse. The first neutron pulse has two different parts. The first part (onset) correlates with the last phase of compression of the dense plasma sheath, transport of the mass in the dense column from the boundary to the z-axis and formation of the plasmoid. The second part (decrease) correlates with the start of expansion of the dense column, with the decrease in the plasma density in the plasmoid and with the transport of the mass from the axis to the column boundary.

The first part of neutron emission (together with pulses of the SXRs and HXRs) started at −30 ns (figure 7(*a*)). Between −48 and −18 ns, the diameter of the dense pinch decreased and the toroidal-like disk structure was compressed to a smaller radius of $r = 1.0$ cm imaged in figure 9(*a*). At this time (−18 ns), the formation of the plasmoidal structure starts. In figure 10, we can see this phase of plasmoid formation in great detail in two other different shots with low and high maximal values of Bz for different distances of the toroidal disk from the probe coil. In both shots in figure 10, we can see the center of the plasmoid at the center of the disk structures, similarly as in shot #9354. The evolution of the plasmoidal structure is imaged in figures 8(*b*) and (*c*) and 9(*a*) and (*b*). The density maximum in the central region of the plasmoids, about (1–2) × 10^{25} m^{-3}, is one order higher than that in the boundaries. When the error associated with the Abel inversion can reach 10–20%, the densities in the central part are considerably higher than those at the boundary. The typical radial and axial Gaussian-like density profiles of plasmoids were imaged in detail in [6] in figure 5. The plasmoid seems to be a source of Bz as well, yet we do not know as to what is the Bz distribution inside it and the distribution of Bz outside the plasmoids. We can only summarize the dependence of Bz measured in the coil on the distance between the coil and the center of the plasmoid z. The value of 0.3 T in shot (#9356)

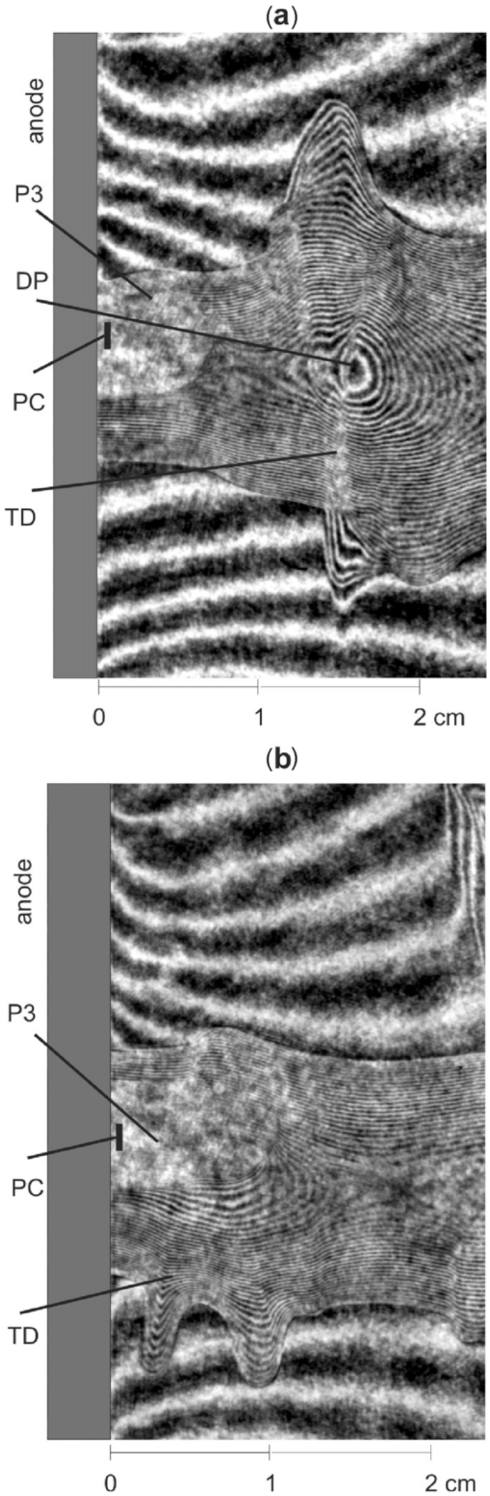

Figure 10. Interferograms registered during the first neutron pulse (*a*) shot #9356 at −1 ns and Bz 0.4 T and (*b*) shot #9357 at −11 ns and Bz 1 T. P3 is the position of the probe body, PC is the position of the probe coil and TD is the z-position of the compressed disk.

was registered at a distance of $z = 1$ cm in figure 9(*a*) and the value of 1 T in shot #9357 at a distance $z = 0.2$–0.5 cm in figure 9(*b*). Then the registered magnetic field is lower at the higher z-distance and Bz in plasmoids could be higher than 1 T.

The second dBz pulse in shot #9354 correlates with the first part of the first neutron pulse (completed at about 0 ns).

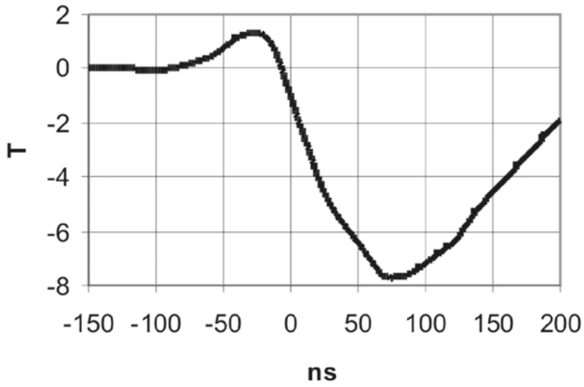

Figure 11. Shot #9357: Bz signal.

The second part of the first neutron pulse, the decrease, in shot #9354 started at about 0 ns, when Bz and the plasma density at the plasmoid center reached their maximum (above 10^{25} m^{-3}). After this time, the Bz value rapidly decreases. In the interferogram registered at +12 ns (figure 9(*b*)), the disk structure is not clear. During the second part of the first neutron pulse, the plasma was transported in the radial direction from the axis to the column boundary.

These two parts of the first neutron pulse are characterized (i) by radial transport of the plasma first to the z-axis and later from the axis to the column boundary, (ii) by the evolution of the plasmoid (figures 9(*a*) and 10(*a*)), first its formation and later its expansion and (iii) by an initial increase in Bz (with the maximal value at the center above 2 T) and its later decrease. Subsequently, the change in Bz correlates with the phase of neutron production.

3.3.3. Evolution of the $m = 0$ instability and the main neutron pulse. In shot #9354, the neutron production continued with the same intensity during the evolution of the constriction from −20 to 100 ns. In figure 9(*b*) registered at 12 ns, we can see stagnation in the form of a dense plasma column characterized by a small variation in density along the axial axis. This is the typical interferometric image registered after the first neutron pulse [6]. The initial phase of the $m = 0$ instability and the variations in the axial distribution of plasma densities are imaged in figure 12(*a*) at 42 ns. This phase starts with the formation of both the neck near the anode and the dense toroidal structure in its downstream neighborhood. The plasma from the imploding neck is injected into the toroidal center, where the axially narrow dense disk structure is created. In figure 12(*b*), registered at 72 ns, we can see a further similar evolution of instabilities in higher z positions—the imploding necks and the expanding dense disks between necks, at the center of which the plasma being injected from the necks is cumulated. The evolution of the toroidal structure and its transformation into the toroidal disk and plasmoid are similar to that described at the final phase of implosion and the start of stagnation. The continuing neutron production during the period registered in figures 12(*a*) and (*b*) correlates with the implosion of the neck and with the increase in the plasma density in the disk near the anode and with the third, negative and limited dBz pulse. The Bz value decreases to zero

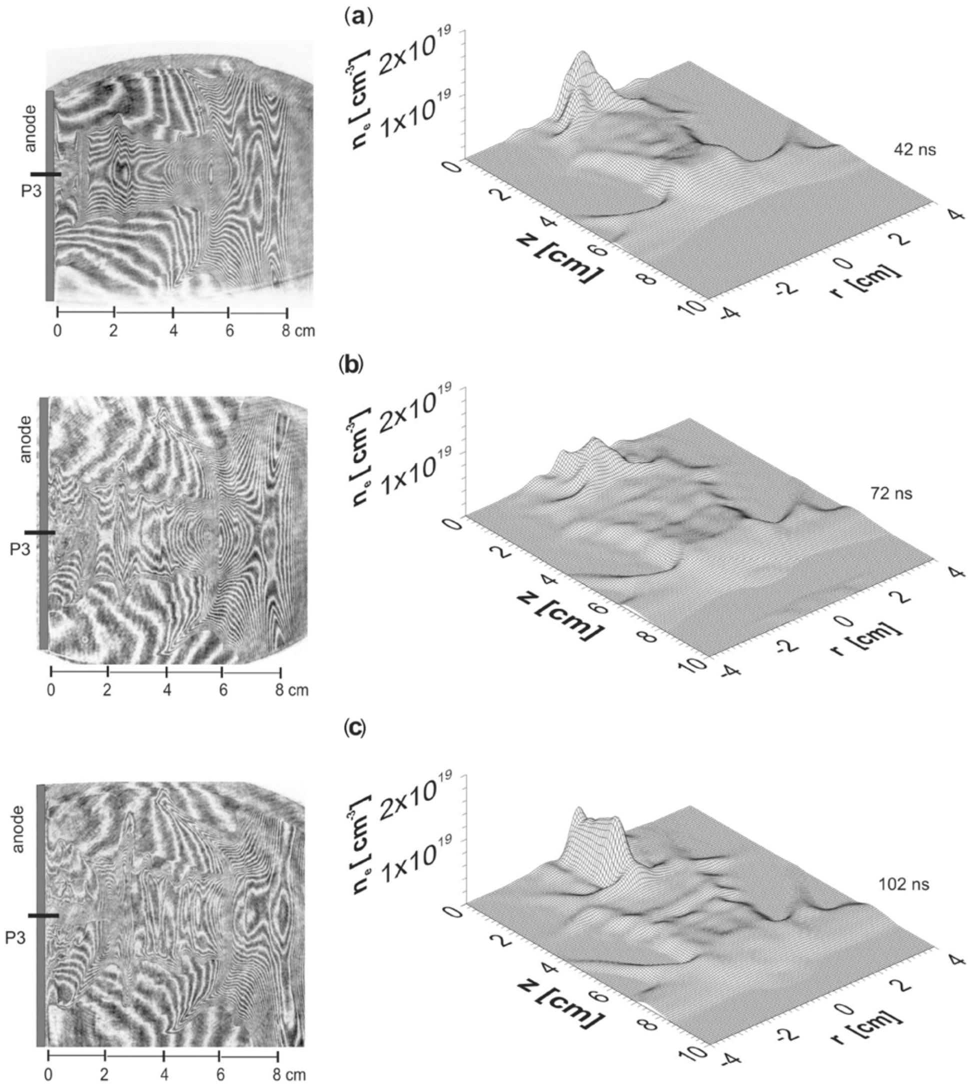

Figure 12. Shot #9354: interferograms and densitograms obtained during evolution of instabilities and formation of the dense constriction and disk structure at (*a*) 42 ns, (*b*) 72 ns and (*c*) 102 ns. P3 is the position of the dBz probe.

probably at about 30–50 ns, and later a Bz with opposite sign (in comparison with Bz registered during implosion and during the first neutron pulse) originates from the long constriction and its neighbor plasmoid. In the constriction, the maximum plasma density of 10^{25} m^{-3} was achieved, as shown in figure 12(*c*).

We can summarize that evolution of the $m = 0$ instability starts in correlation with the tearing of the component of the Bz field along the z-axis. The dense toroidal structures between constrictions can be a source of Bz in analogy with the one registered during the implosion phase. Their Bz value increases during evolution of the dense plasmoidal disks, while the necks can implode due to the lower initial internal Bz. The value of Bz in the plasmoidal structure reaches its maximum closure before its explosion (in correlation with the intense pulse of neutron production).

The value of Bz inside the plasmoid 20 ns before its explosion and intense neutron production can be evaluated from shot #9357 at 49 ns (figure 13), when Bz reached a value of 6 T (figure 11). In figure 13 we see that the plasmoid is spread around the body of the probe in this shot at this time.

4. Summary

The plasma implosion and pinch phase of a plasma focus discharge were studied using magnetic probes placed in the anode front at radial distances of 0, 1.3 and 4 cm at the current in the pinch of up to 1.5 MA. The probes registered the temporal derivative of both axial dBz and azimuthal d$B\varphi$ components of magnetic inductance, which were correlated with interferometry figures and neutron signals.

The probes positioned at a distance of 4 cm allow description of the temporal current distribution of the imploded current sheath. The current sheath imploding the dense plasma layer contains $80 \pm 20\%$ of the total discharge current. Knowing the implosion velocity, we can estimate the width of the current sheath to be about 1.6–2.6 cm. The current passes outside the dense plasma layer. The current sheath is composed of both $B\varphi$ and Bz components, in which the amplitude of the Bz component is roughly 2–8 times lower. In front of this sheath, a layer with a weak Bz prepulse was registered with the opposite sign in comparison with the orientation observed in the current sheath.

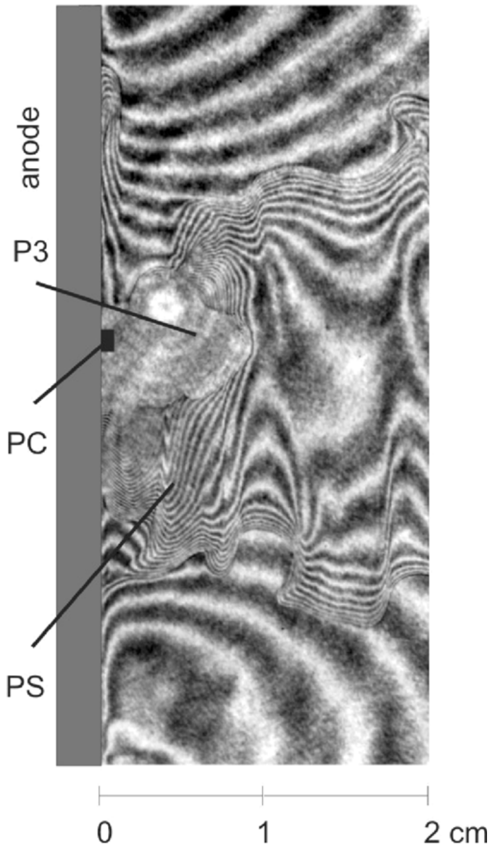

Figure 13. Shot #9357: interferogram registered at 49 ns in the time close before the dominant neutron pulse and at Bz maximum, when the probe coil PC in the probe body P3 registered 6 T.

The d$B\varphi$ waveforms registered with probes positioned at 1.3 cm distance confirmed the continuation of the motion of the current sheath to the z-axis after reaching the minimal diameter at the stop of implosion of the plasma column. Therefore, the current sheath penetrates into the dense plasma column during the first neutron pulse. At the final phase of stagnation, the main part of the discharge current passes through this dense column.

The first neutron pulse is generated in correlation with the formation (usually at the center of the toroidal structures) and disintegration of dense plasmoids, with first an increase and later a decrease in Bz registered at the probe positioned on the anode axis. The signals registered at this position also described the variation of the axial magnetic field during the evolution of the $m = 0$ instability. Bz increases in the regions with dense disks between necks. During evolution of the neck, the plasma is ejected from the neck to the center of the disk, where the plasma density increases and the plasmoid is formed. During the dominant neutron production, at the time of the explosion of the constriction and plasmoid, Bz decreases.

Under the supposition of the existence of Bz in the toroidal and plasmoidal structures, a maximum of about 2 T was reached inside the plasmoid correlated with the first neutron pulse and above 6 T inside the plasmoids formed close before the main neutron pulse. The sign of Bz registered during the first neutron pulse is the same as in the prepulse of the current sheath, but opposite to the sign registered in the current sheath. At this time, $B\varphi$ on the boundary of the plasma column reaches a value of 15–25 T (estimated for the total current inside a diameter of 1.0–2.0 cm). Therefore, the plasmoids and toroidal disks can be composed of closed currents with the axial component of magnetic fields. The change in magnetic field during plasmoid formation and disintegration, correlated with XR and neutron production, confirms the existence of the reconnection of magnetic lines as one of the possible mechanisms for high-energy particle acceleration.

The estimated values of magnetic fields of 5–20 T, dimensions of plasmoids of cm and the velocity of transformation equal to the Alfvén velocity of (1 –2) $\times$ 10^5 m s^{-1} are insufficient for the generation of hundreds of kV voltage necessary for fast deuteron acceleration. Nonetheless, they can be a base for the consideration of pinch plasmas assuming a filamentary structure for current of dimension tens and hundreds of μm and with magnetic fields of hundred teslas in more detail [15, 16].

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